LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control surface and tabs, aerodynamics characteristics of trailing edge controls, types of tabs, all moving surface, mechanical and structural aspects of control surface design
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control surface and tabs, aerodynamics characteristics of trailing edge controls, types of tabs, all moving surface, mechanical and structural aspects of control surface design
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of control surface cutaway models; verification of hinge line alignment and tab actuation linkage; safety: Handle sharp edges with care; Instructor supervision required during model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft wing?
(b) Define the function of a trailing edge control surface.
(c) List the three main types of trailing edge controls.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and function of control surfaces • Aerodynamic characteristics of trailing edge controls • Types of tabs: Anti-servo, Balance, Friction • All-moving surface design and hinge moment • Mechanical linkage systems for control surfaces	L & Ds	PPT, IB	15 min
(b)	• Structural elements of control surface design • Material selection and skin thickness requirements • Rib and spar integration in control surfaces • Hinge mechanisms and pivot points • Load paths and stress distribution analysis	L & Ds	PPT, PDF	20 min
(c)	• Failure modes of control surface structures • Inspection criteria for hinges and linkages • Maintenance procedures for control surface systems • Troubleshooting common structural defects • Integration with primary wing structure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the aerodynamic purpose of an anti-servo tab.
(b) Describe the structural load path from the control surface to the wing spar.
(c) Identify two critical inspection points for control surface hinges.

12. Home Assignment and Follow Up	Prepare a brief note comparing the structural design requirements of ailerons versus elevators, referencing the next lesson on aero-elasticity.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to aero-elasticity, classification, effects of structural distortion, wing torsional divergence, control effective and reversal. introduction to flutter, control surface flutter, wing bending torsional flutter, factor effecting flutter characteristics, aileron buzz, buffeting, dynamic response, effect of sweep on various critical speeds	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min